Lecture 10 — Mock National Championship & Post-Mortem

> National Championship Prep · Prerequisites: Nationals Lectures 1–9 (and the Year Curriculum and Regional track) · Estimated time: 5 h contest + 1.5 h debrief, and a 1 h post-mortem presentation in the following session

1. Overview

This session is a full simulation of a national championship (ICPC North America Championship, a national olympiad final, or a Codeforces Div 1 round stretched to five hours). The problem set below is **original** and was written for this course. Every problem is solvable with techniques from this track and its prerequisites, but the statements never name the technique, and — as in real contests — the problems are **not** ordered by difficulty.

The second half of the lecture is about the **post-mortem**. Teams that improve between regionals and nationals are rarely the ones that learn the most new algorithms; they are the ones that systematically find out *why* they lost points and change one habit at a time. Section 4 gives a template you should fill in after this mock and every future one.

How to use these notes:

- **Before the contest**: students see only the problem statements handout (Section 3's "Statement" paragraphs, expanded with full I/O format by the instructor).
- **During the debrief**: go through Section 3 in the order of the expected difficulty curve (A, E, F, C, K, B, H, J, D, G, I, L).
- **Afterwards**: upsolve every problem. The solutions are held by the coach and released only after the contest has been run; they are not part of these notes.

2. Contest rules

- Teams of three, one computer, five hours, a printed team reference document (≤ 25 pages, made in Lecture 1).
- ICPC scoring: number solved, then penalty time = Σ over solved problems of (minutes until AC + 20 × rejected submissions on that problem).
- Scoreboard freezes for the last hour.
- Language: C++17 with `-O2`. Default 8 MB stack — deep recursion on $3 \cdot 10^5$-vertex graphs will crash; plan for it.

- All input is integral unless stated; output real numbers with absolute or relative error ≤ 10^{-6}.

3. The problem set

Ratings are estimates on the Codeforces scale.

	Problem	Rating
A	Lanterns	1300
B	Recurring Revenue	2100
C	Toll Bridges	2000
D	Distance Band	2300
E	Echo	1900
F	Corridor	1900
G	Overtime Assignment	2300
H	Twin Stations	2200
I	Xor Treasure Map	2300
J	Dice Tower	2200
K	Lighthouse	2000
L	Warehouse Districts	2400

> **Solutions are not published here.** The coach holds the solutions and releases them only after the contest has been run.

Only the statements are given below; solution sketches, traps and reference code are not published.

A — Lanterns (≈1300)

Statement. A street is the segment $[0, L]$ ($1 \le L \le 10^9$). There are $n \le 2 \cdot 10^5$ lanterns; lantern i at integer position p_i with radius r_i lights the closed segment $[p_i - r_i, p_i + r_i]$. Output the minimum number of lanterns to switch on so that every point of $[0, L]$ is lit, or -1.

B — Recurring Revenue (≈2100)

Statement. Given $k \le 20$, coefficients $c_1..c_k$, constants d, e, and $a_0..a_{k-1}$. For $i \ge k$, $a_i = \sum_j c_j a_{i-j} + d \cdot i + e \pmod{10^9+7}$. Answer $q \le 2000$ queries: a_n for $n \le 10^{18}$.

C — Toll Bridges (≈2000)

Statement. A connected undirected graph with n, m ≤ $3 \cdot 10^5$ (multi-edges allowed, no self-loops). q ≤ $3 \cdot 10^5$ queries (u, v): how many edges are there whose removal (alone) disconnects u from v?

D — Distance Band (≈2300)

Statement. An unweighted tree with n ≤ $2 \cdot 10^5$ vertices and integers $1 \le L \le R \le n$. Count unordered pairs {u, v}, $u \ne v$, with $L \le \text{dist}(u, v) \le R$.

E — Echo (≈1900)

Statement. A lowercase string s, n ≤ $2 \cdot 10^5$, and an integer $1 \le k \le n$. Output the maximum length of a substring that occurs at least k times in s (occurrences may overlap); output 0 if none.

F — Corridor (≈1900)

Statement. n ≤ $2 \cdot 10^5$ integer points, |coordinates| ≤ 10^9. Find the minimum width of an infinite straight strip (region between two parallel lines) containing all points.

G — Overtime Assignment (≈2300)

Statement. n ≤ 100 workers and m ≤ 100 tasks. Worker i can do task j at cost $c_{ij} \ge 0$ (or not at all). Every task must be done by exactly one worker. A worker who does t tasks additionally pays $o_i \cdot t(t-1)/2$ (the k-th task costs $o_i \cdot (k-1)$ extra). Minimize the total, or report impossible.

H — Twin Stations (≈2200)

Statement. n ≤ 1000 stations, 2 ≤ n. Station i must be built at position a_i or b_i on a line (integers, $|\cdot| \le 10^9$). Maximize the minimum distance between two built stations.

I — Xor Treasure Map (≈2300)

Statement. A connected undirected graph, n, m, q ≤ $2 \cdot 10^5$, weights $0 \le w < 2^{60}$ (self-loops and multi-edges allowed). The value of a walk is the xor of the weights of the edges it traverses, counted with multiplicity. For each query (u, v) output the maximum value of a walk from u to v.

J — Dice Tower (≈2200)

Statement. n dice; die i has f_i faces with values $v_{i1}..v_{i_\{f_i\}} \in [0, 10^5]$ (values may repeat). $\Sigma f_i \le 2 \cdot 10^5$ and $S = \Sigma_i \max_j v_{ij} \le 2 \cdot 10^5$. For every t in [0, S], output the number of ways (one face per die) to get total t, mod 998244353.

K — Lighthouse (≈2000)

Statement. A lighthouse at the origin; $n \le 2 \cdot 10^5$ ships at integer points (not the origin, $|\text{coordinates}| \le 10^9$) with values $1 \le w_i \le 10^9$. The beam is a closed sector with apex at the origin, infinite radius, and angle α degrees (integer, $1 \le \alpha \le 359$). Output the maximum total value of ships inside one beam position. It is guaranteed that the answer does not change if α changes by 10^{-6} degrees.

L — Warehouse Districts (≈2400)

Statement. $n \le 10^5$ positive integers $a_i \le 10^4$ and $k \le \min(n, 100)$. Split the array into exactly k non-empty contiguous districts minimizing Σ (district sum)².

4. The post-mortem

4.1 Immediately after the contest (30 minutes, before the solutions are shown)

Write the **raw timeline** while memory is fresh: for each event, the time, the person, and what happened ("1:07 Bob reads K, labels it medium", "1:31 C WA on test 14", "2:10 keyboard idle 9 minutes"). Export the submission log from the judge. Do not analyse yet.

4.2 Per-problem table

Fill in one row per problem, including problems you never opened.

Problem	Status (AC time / attempts)	First read (who, when)	Idea found (when)	Coding time	Wrong runs and cause	Root cause category	Note
A							
B							
…							
L							

Root-cause categories (choose one primary per lost problem or wasted interval):

1. **Reading** — misread constraint/output format; missed a guarantee.
2. **Idea** — did not find the algorithm; or found a wrong one and did not prove it.
3. **Knowledge** — knew the idea but not the technique (e.g., had never coded the data structure the idea needed).
4. **Implementation** — bug in code that was conceptually right.
5. **Edge case / limits** — overflow, n = 1, recursion depth, precision.
6. **Strategy** — wrong problem order, two people on one problem, sunk-cost on a hard problem, idle keyboard.

4.3 Team-level questions

- Which problems did we never read carefully, and how hard were they really?
- When was the keyboard idle, and why?
- For each WA: would a stress test or a careful re-read of the statement have caught it? How long would that have taken?
- **Counterfactual score:** using only ideas the team *had* during the contest, what was the best achievable result with perfect execution? The gap between this and the real score is the strategy/implementation gap; the gap to the full set is the knowledge/idea gap. They need different training.
- What did the scoreboard tell us at 1:00 and 2:30 that we ignored?

4.4 Action items

At most **three**, each concrete and checkable at the next mock. Good: "Re-read every unsolved statement at the 2:00 mark", "Add an iterative DFS template to the notebook and test it on a 10^6 path", "Whoever gets the second WA on a problem writes a brute force before resubmitting". Bad: "Be more careful", "Practise more geometry".

4.5 Individual reflection (each member, 10 minutes)

- One problem I should have solved and why I did not.
- One technique to drill this week (with 3 concrete practice problems).
- One habit to keep.

4.6 Notebook diff

List every piece of code you wished you had: missing templates (e.g., anything someone had to write from scratch during the contest), wrong ones, and ones that were too hard to find. Update and re-print the notebook within the week.

5. Common mistakes in mock contests

- **Reading in letter order.** In this set E, F, C and K are among the five easiest; teams that read linearly often find them after B and D have eaten an hour.
- **No division of the set.** Split the problems among the three members for the first read, then report one-line summaries.
- **Coding before the complexity check.** A solution that "works" on the samples can still TLE; estimate the worst-case operation count, not just the big-O, before typing.
- **Silent duplication.** Two people solving the same problem in their heads for 40 minutes.
- **No brute force after repeated WA.** The second wrong answer on a problem should trigger a stress test (Year week 35).
- **Treating the mock as a test, not a measurement.** The score matters far less than the post-mortem; teams that skip Section 4 do not improve.

6. Practice problems

Full sets for the next simulations (pick ones no team member has seen):

- ICPC North America Championship problem sets — available on Kattis (search the problem source "ICPC North America Championship") — https://open.kattis.com/
- ICPC Northern Eurasia (NERC) finals on Codeforces, e.g. 2023–2024 NERC — https://codeforces.com/contest/1912
- NWERC, SWERC, CERC sets in the Codeforces Gym — https://codeforces.com/gyms

7. Further reading

- Lecture 1 of this track (training system, team roles, notebook) — re-read it before writing your post-mortem.
- Year week 35 (contest strategy, stress testing) and Regional 01 (contest anatomy, triage).
- ICPC World Finals problem analyses and NAC editorials published by the judges each year — compare your post-mortem categories with what the judges expected to be hard.
- Next: World Finals track, Lecture 1 — WF format, problem archetypes, and five-hour strategy.